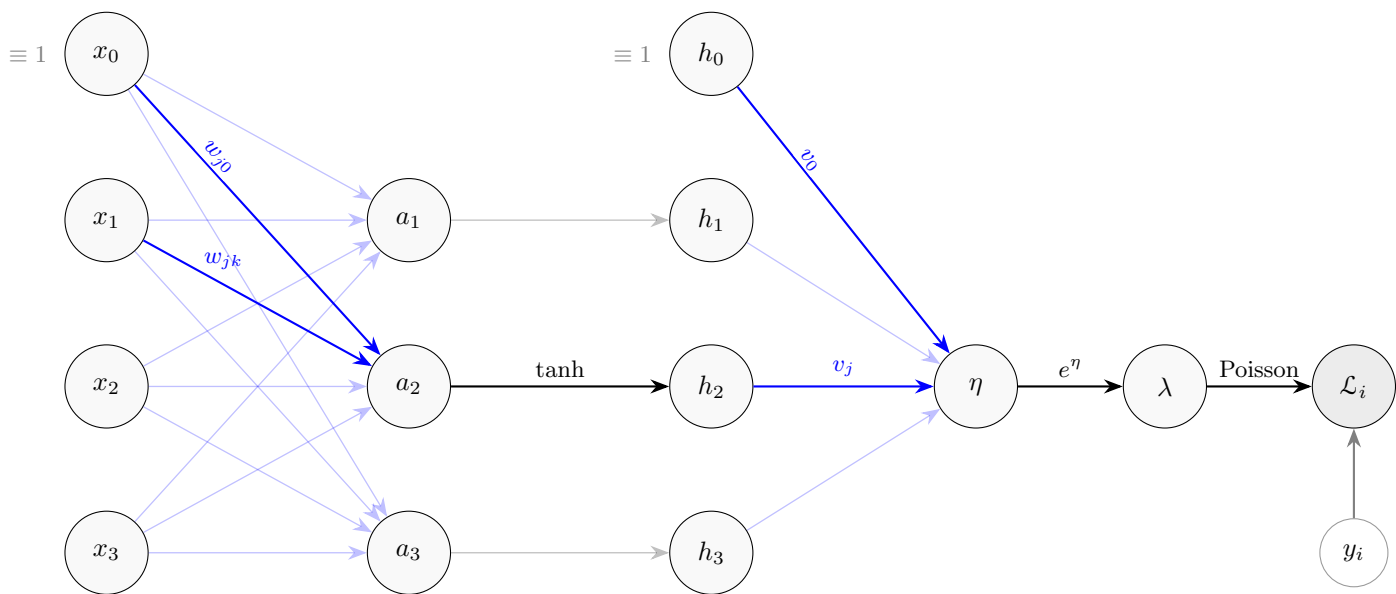
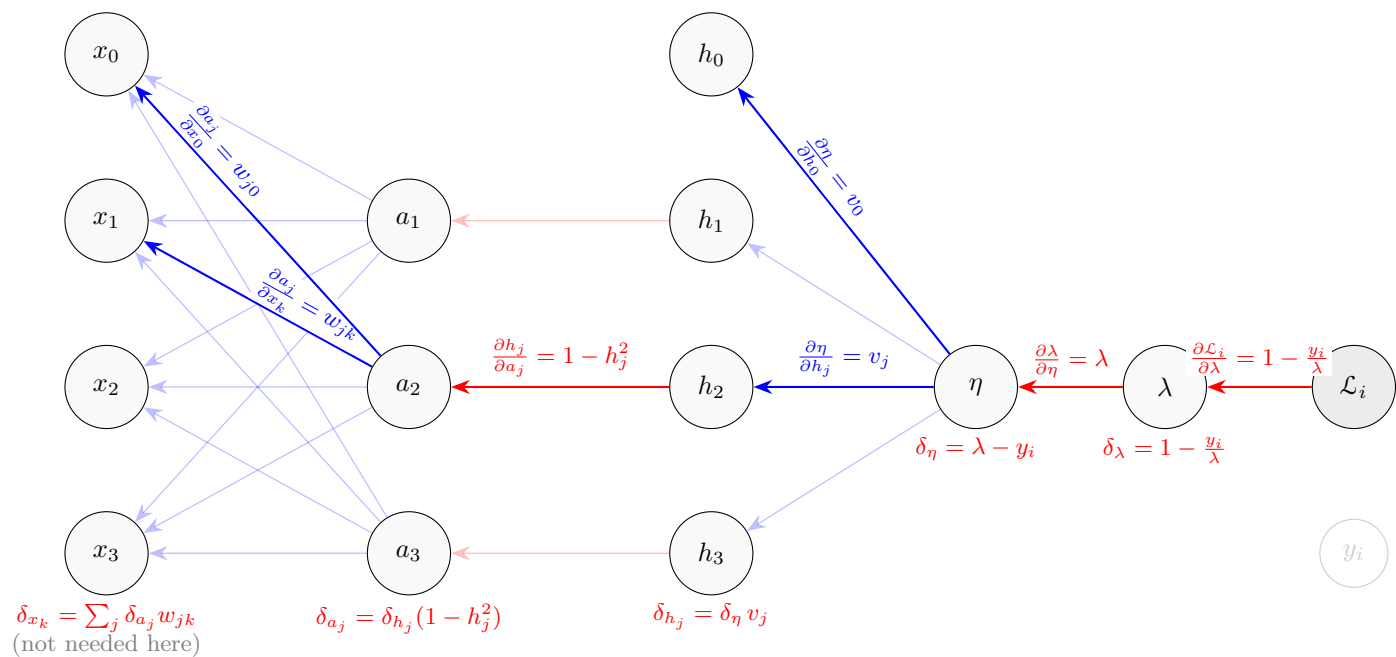


forward pass: values



backward pass: local derivatives



$$\frac{\partial \mathcal{L}_i}{\partial w_{jk}} = \delta_{a_j} x_k$$

$$\frac{\partial \mathcal{L}_i}{\partial w_{j0}} = \delta_{a_j} x_0 = \delta_{a_j}$$

$$\frac{\partial \mathcal{L}_i}{\partial v_j} = \delta_\eta h_j$$

$$\frac{\partial \mathcal{L}_i}{\partial v_0} = \delta_\eta h_0 = \delta_\eta$$